Magnetic Effect of a Current

Exam code: 0625 & 0972

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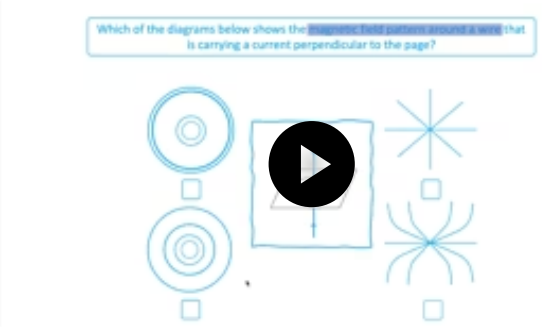

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Magnetic fields around wires & solenoids

- **Magnetic fields** are formed wherever a current flows, such as in:
  - straight wires
  - solenoids
  - circular coils

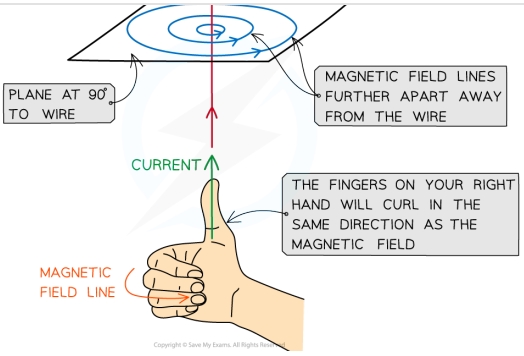
Magnetic field due to a straight wire

- The magnetic field lines around a straight wire are
  - made up of **concentric circles**
  - centred on the wire
- A circular field pattern indicates that the magnetic field around a current-carrying wire has **no poles**
- The **right-hand grip rule** can be used to work out the **direction** of the magnetic field

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



The direction of the field around a current-carrying wire can be determined using the right-hand grip rule

- The field lines are clockwise or anticlockwise around the wire, depending on the direction of the current
  - Reversing the current reverses the direction of the field

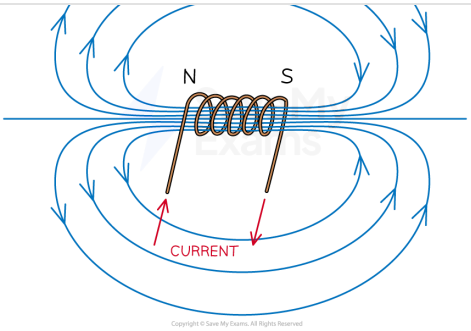
- The direction of the magnetic field can be determined using the **right-hand grip rule**
  - This is determined by pointing the **right-hand** thumb in the direction of the current in the wire and curling the fingers onto the palm
  - The direction of the curled fingers represents the direction of the magnetic field lines around the wire
  - For example, if the current is travelling vertically upwards, the magnetic field lines will be directed anticlockwise, as seen from directly above the wire

- **Note:** the direction of the current is taken to be the conventional current i.e. from **positive** to **negative**, **not** the direction of electron flow

Magnetic field due to a solenoid

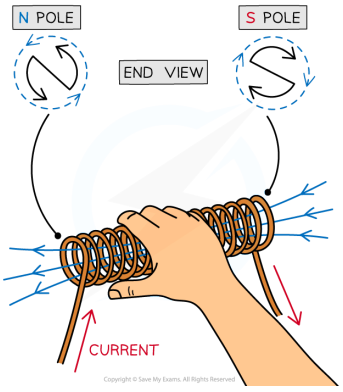
- As seen from a current-carrying wire, an electric current produces a magnetic field
- An electromagnet utilises this by using a coil of wire called a **solenoid**
  - This increases the strength of the magnetic field by adding more **turns** of wire into a smaller region of space

- One end of the solenoid becomes a north pole, and the other becomes the south pole



The magnetic field lines around a solenoid are similar to a bar magnet

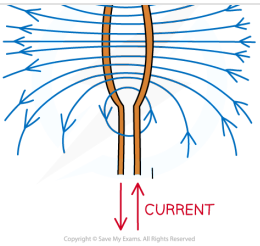
- As a result, the field lines around a solenoid are similar to those around a bar magnet
  - The field lines **emerge** from the **north** pole
  - The field lines **return** to the **south** pole
- The poles of the solenoid can be determined using the **right-hand grip rule**
  - The curled fingers represent the direction of the current flow around the coil
  - The thumb points in the direction of the field inside the coil, towards the **north pole**



In a solenoid, the north pole forms at the end where the current flows anti-clockwise, and the south pole at the end where the current flows clockwise

Magnetic field due to a circular coil

- A circular coil is equivalent to one of the coils of a solenoid
- The field lines emerge through one side of the circle (north pole) and enter through the other (south pole)
- As with a solenoid, the direction of the magnetic field lines depends on the direction of the current
  - This can also be determined using the **right-hand grip rule**



Magnetic field lines of many individual circular coils can be combined to make a solenoid

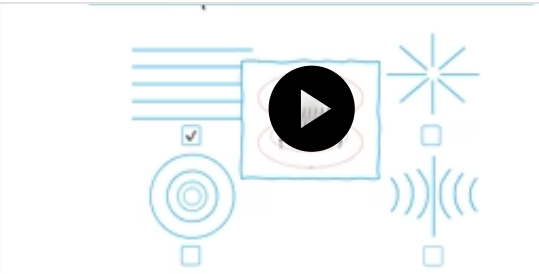
Worked Example

The current in a long, straight vertical wire is in the direction XY, as shown in the diagram.

Sketch the magnetic field lines in the horizontal plane ABCD due to the current-carrying wire. Draw at least four field lines.

Answer:

- Concentric circles
- Increasing separation between each circle
- Arrows drawn in an anticlockwise direction



Did this video help you? Yes No

Magnetic effects of changing current

Extended tier only

Magnetic field strength around a straight wire

- The strength of the magnetic field around a wire depends on:
  - the size of the **current**
  - the **distance** from the wire
- The strength of a magnetic field **increases** as the amount of **current** flowing through the wire increases
  - This means the field lines will become **closer together**
- The strength of a magnetic field **decreases** with **distance** from the wire
  - The magnetic field is strongest near the wire and becomes weaker further away from the wire
  - This is shown by the magnetic field lines becoming **further apart**
- When the **direction** of the current changes, the magnetic field acts in the **opposite** direction

The greater the current, the stronger the magnetic field. This is shown by more concentrated field lines

- The strength of the magnetic field produced around a **solenoid** can be increased by:
  - increasing the amount of **current** flowing through the coil
  - increasing the **number of turns** on the coil
  - inserting an **iron core** into the coil

- When a soft **iron core** is inserted into a solenoid, it can be used as an electromagnet
- The iron core becomes an **induced magnet** when a current flows through the coils
- The magnetic field produced by the solenoid and the iron core will create a much **stronger** magnet overall

An electromagnet consists of a solenoid wrapped around a soft iron core

- Changing the direction of the current also changes the direction of the magnetic field produced by the iron core

Applications of the magnetic effect of a current

- Electromagnets are used in a wide variety of applications, including:
  - relay circuits (utilised in electric bells, electronic locks, scrapyard cranes etc)
  - loudspeakers

Relay circuits

- Electromagnets are commonly used in **relay circuits**
- Relays are switches that open and close via the action of an electromagnet
- A relay circuit consists of:
  - an electrical circuit containing an electromagnet
  - a second circuit with a switch which is near to the electromagnet in the first circuit

Operation of a relay circuit

When a current passes through the coil in Circuit 1, it attracts the switch in Circuit 2, and closing it enables a current to flow in Circuit 2

- When a current flows through **Circuit 1**:
  - a magnetic field is induced around the coil
  - the magnetic field attracts the switch, causing it to pivot and close the contacts in **Circuit 2**
  - this allows a current to flow in **Circuit 2**
- When no current flows through **Circuit 1**:
  - the magnetic force stops
  - the electromagnet stops attracting the switch
  - the current in **Circuit 2** stops flowing
- Scrapyard cranes utilise relay circuits to function:
  - When the electromagnet is switched on, it will **attract** magnetic materials
  - When the electromagnet is switched off, it will **drop** the magnetic materials
- Electric bells also utilise relay circuits to function:

Animation: Electric bells utilise relay circuits. As the current alternates, the metal arm strikes the bell and drops repeatedly to produce the ringing effect

- When the button **K** is pressed:
  - a current passes through the electromagnet **E** creating a magnetic field
  - this attracted the iron armature **A**, causing the hammer to strike the bell **B**
  - the movement of the armature breaks the circuit at **T**
  - this stops the current, destroying the magnetic field and so the armature returns to its previous position
  - this re-establishes the circuit, and the whole process starts again

Loudspeakers

- A loudspeaker consists of a coil of wire which is wrapped around one pole of a **permanent magnet**

Structure of a loudspeaker

A loudspeaker converts the a.c. of an electrical signal into sound waves

- An **alternating current** passes through the coil of the loudspeaker
  - This creates a **changing magnetic field** around the coil
- As the current is constantly changing direction, the direction of the magnetic field will be **constantly changing**
- The magnetic field produced around the coil **interacts** with the field from the permanent magnet
- The interacting magnetic fields will exert a **force** on the coil
- As the magnetic field is constantly changing direction, the force exerted on the coil will **constantly change direction**
  - This makes the coil **oscillate**
- The oscillating coil causes the speaker cone to oscillate
  - This makes the air oscillate, creating **sound waves**

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